

Problem 1

Solution:

Graphs are omitted. Use <https://www.desmos.com/> instead

(a)

$$h(x) = \begin{cases} -3 & \text{if } -1 \leq x < -\frac{1}{2} \\ -2 & \text{if } -\frac{1}{2} \leq x < 0 \\ 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x < 1 \\ 3 & \text{if } 1 \leq x < \frac{3}{2} \\ 4 & \text{if } \frac{3}{2} \leq x < 2 \\ 6 & \text{if } x = 2 \end{cases}$$

(b)

$$h(x) = \begin{cases} -2 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } 0 \leq x < 1 \\ 2 & \text{if } 1 \leq x < 2 \\ 4 & \text{if } x = 2 \end{cases}$$

(c)

$$h(x) = \begin{cases} 2 & \text{if } -1 \leq x < -\frac{1}{2} \\ 1 & \text{if } -\frac{1}{2} \leq x < 0 \\ 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 0 & \text{if } \frac{1}{2} \leq x < 1 \\ 2 & \text{if } 1 \leq x < \frac{3}{2} \\ 3 & \text{if } \frac{3}{2} \leq x < 2 \\ 8 & \text{if } x = 2 \end{cases}$$

(d)

$$h(x) = \begin{cases} \frac{1}{2} & \text{if } -1 \leq x < -\frac{1}{2} \\ \frac{1}{4} & \text{if } -\frac{1}{2} \leq x < 0 \\ 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 0 & \text{if } \frac{1}{2} \leq x < 1 \\ \frac{1}{2} & \text{if } 1 \leq x < \frac{3}{2} \\ \frac{3}{4} & \text{if } \frac{3}{2} \leq x < 2 \\ 2 & \text{if } x = 2 \end{cases}$$

□

Problem 2

Solution:

Graphs are omitted. Use <https://www.desmos.com/> instead

(a) Not a step function over any interval of real x

$$f(x) = \begin{cases} x - 2 & \text{if } -2 \leq x < -1 \\ x - 1 & \text{if } -1 \leq x < 0 \\ x & \text{if } 0 \leq x < 1 \\ x + 1 & \text{if } 1 \leq x < 2 \\ 4 & \text{if } x = 2 \end{cases}$$

(b) Not a step function over any interval of real x

$$f(x) = \begin{cases} x + 2 & \text{if } -2 \leq x < -1 \\ x + 1 & \text{if } -1 \leq x < 0 \\ x & \text{if } 0 \leq x < 1 \\ x - 1 & \text{if } 1 \leq x < 2 \\ 0 & \text{if } x = 2 \end{cases}$$

(c) A partition over which f is a step function is $P = \{-2, -1, 0, 1, 2\}$.

$$f(x) = \begin{cases} 2 & \text{if } x = -2 \\ 1 & \text{if } -2 < x \leq -1 \\ 0 & \text{if } -1 < x \leq 0 \\ -1 & \text{if } 0 < x \leq 1 \\ -2 & \text{if } 1 < x \leq 2 \end{cases}$$

(d) A partition over which f is a step function is $P = \{-2, -1, 0, 1, 2\}$.

$$f(x) = \begin{cases} -4 & \text{if } -2 \leq x < -1 \\ -2 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } 0 \leq x < 1 \\ 2 & \text{if } 1 \leq x < 2 \\ 4 & \text{if } x = 2 \end{cases}$$

(e) A partition over which f is a step function is $P = \{-2, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, 2\}$.

$$f(x) = \begin{cases} -2 & \text{if } -2 \leq x < -\frac{3}{2} \\ -1 & \text{if } -\frac{3}{2} \leq x < -\frac{1}{2} \\ 0 & \text{if } -\frac{1}{2} \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x < \frac{3}{2} \\ 2 & \text{if } \frac{3}{2} \leq x \leq 2 \end{cases}$$

(f) A partition over which f is a step function is $P = \{-2, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$.

$$f(x) = \begin{cases} -4 & \text{if } -2 \leq x < -\frac{3}{2} \\ -3 & \text{if } -\frac{3}{2} \leq x < -1 \\ -2 & \text{if } -1 \leq x < -\frac{1}{2} \\ -1 & \text{if } -\frac{1}{2} \leq x < 0 \\ 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x < 1 \\ 2 & \text{if } 1 \leq x < \frac{3}{2} \\ 3 & \text{if } \frac{3}{2} \leq x < 2 \\ 4 & \text{if } x = 2 \end{cases}$$

□

Problem 3

Solution:

Graphs are omitted. Use <https://www.desmos.com/> instead

(a)

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 4 \\ 2 & \text{if } 4 \leq x < 9 \\ 3 & \text{if } 9 \leq x \leq 10 \end{cases}$$

(b)

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < \sqrt{2} \\ 2 & \text{if } \sqrt{2} \leq x < \sqrt{3} \\ 3 & \text{if } \sqrt{3} \leq x < 2 \\ 4 & \text{if } 2 \leq x < \sqrt{5} \\ 5 & \text{if } \sqrt{5} \leq x < \sqrt{6} \\ 6 & \text{if } \sqrt{6} \leq x < \sqrt{7} \\ 7 & \text{if } \sqrt{7} \leq x < \sqrt{8} \\ 8 & \text{if } \sqrt{8} \leq x < 3 \\ 9 & \text{if } x = 3 \end{cases}$$

(c)

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 2 \\ \sqrt{2} & \text{if } 2 \leq x < 3 \\ \sqrt{3} & \text{if } 3 \leq x < 4 \\ 2 & \text{if } 4 \leq x < 5 \\ \sqrt{5} & \text{if } 5 \leq x < 6 \\ \sqrt{6} & \text{if } 6 \leq x < 7 \\ \sqrt{7} & \text{if } 7 \leq x < 8 \\ 2\sqrt{2} & \text{if } 8 \leq x < 9 \\ 3 & \text{if } 9 \leq x < 10 \\ \sqrt{10} & \text{if } x = 10 \end{cases}$$

(d)

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 2 \\ 4 & \text{if } 2 \leq x < 3 \\ 9 & \text{if } x = 3 \end{cases}$$

□

Problem 4

Solution:

(a) From an exercise in the introduction part of the text, we know that for any real x , there is exactly one integer p such that $p \leq x < p + 1$. We call this p the greatest integer of x or $[x]$. Now, observe that if n is any arbitrary integer, $p + n \leq x + n < (p + n) + 1$ where $p + n$ and $(p + n) + 1$ are integers. But this means that $p + n = [x + n]$. Thus, substituting for p , we see that $[x] + n = [x + n]$.

(b) From an exercise in the introduction part of the text, we know that for any real x , $[x] \leq x < [x] + 1$ where $[x]$ is unique. There are two cases. If x is an integer $[x] = x$ which implies that $-[x] = -x$ and in turn, $-[x] \leq -x < -[x] + 1$. Therefore, $-[x] = [-x]$. Alternatively, if $[x] \neq x$ the first inequality tells us $[x] < x < [x] + 1$. This implies $-[x] - 1 < -x < -[x]$, i.e. that $-[x] - 1 = [-x]$.

(c) From an exercise in the introduction part of the text, we know that for any real x, y , $[x] \leq x < [x] + 1$ and $[y] \leq y < [y] + 1$ where $[x], [y]$ are unique. Therefore,

$$[x] + [y] \leq x + y < [x] + [y] + 2$$

Now consider $[x] + [y] + 1$. There are two cases.

If $[x] + [y] + 1 \leq x + y$ then we have $[x] + [y] + 1 \leq x + y < [x] + [y] + 2$, implying that $[x + y] = [x] + [y] + 1$. And if $[x] + [y] + 1 > x + y$ we have $[x] + [y] < x + y < [x] + [y] + 1$, implying that $[x + y] = [x] + [y]$.

Thus, we can conclude the proposition and also refine it

$$[x + y] = \begin{cases} [x] + [y] & \text{if } x + y < [x] + [y] + 1 \\ [x] + [y] + 1 & \text{if } x + y \geq [x] + [y] + 1 \end{cases}$$

(d) From part (c) we know that

$$\left[x + \frac{1}{2} \right] = \begin{cases} [x] & \text{if } x + \frac{1}{2} < [x] + 1 \implies x < [x] + \frac{1}{2} \\ [x] + 1 & \text{if } x + \frac{1}{2} \geq [x] + 1 \implies x \geq [x] + \frac{1}{2} \end{cases}$$

since $[\frac{1}{2}] = 0$.

Suppose the first condition, i.e. $x < [x] + \frac{1}{2}$ holds. Then, $[x + \frac{1}{2}] = [x]$, which implies that $[x] + [x + \frac{1}{2}] = [x] + [x]$. Also, observe that $x < [x] + \frac{1}{2} \implies 2x < 2[x] + 1 \implies x + x < [x] + [x] + 1$. Hence, $[2x] = [x] + [x]$ so we can conclude that $[2x] = [x] + [x + \frac{1}{2}]$.

Alternatively, suppose that $x \geq [x] + \frac{1}{2}$. Then, $[x + \frac{1}{2}] = [x] + 1 \implies [x] + [x + \frac{1}{2}] = [x] + [x] + 1$. Also, observe that $x \geq [x] + \frac{1}{2} \implies 2x \geq 2[x] + 1 \implies x + x \geq [x] + [x] + 1$. Hence, $[2x] = [x] + [x] + 1$ so we can conclude that $[2x] = [x] + [x + \frac{1}{2}]$.

(e) From part (c) (along with part (d) in the third part) we know that

$$\left\lceil x + \frac{1}{3} \right\rceil = \begin{cases} [x] & \text{if } x + \frac{1}{3} < [x] + 1 \implies x < [x] + \frac{2}{3} \\ [x] + 1 & \text{if } x + \frac{1}{3} \geq [x] + 1 \implies x \geq [x] + \frac{2}{3} \end{cases}$$

and

$$\left\lceil x + \frac{2}{3} \right\rceil = \begin{cases} [x] & \text{if } x + \frac{2}{3} < [x] + 1 \implies x < [x] + \frac{1}{3} \\ [x] + 1 & \text{if } x + \frac{2}{3} \geq [x] + 1 \implies x \geq [x] + \frac{1}{3} \end{cases}$$

and

$$[3x] = [x + 2x] = \begin{cases} [x] + [2x] = 2[x] + \left\lceil x + \frac{1}{2} \right\rceil & \text{if } 3x < 2[x] + \left\lceil x + \frac{1}{2} \right\rceil + 1 \\ [x] + [2x] + 1 = 2[x] + \left\lceil x + \frac{1}{2} \right\rceil + 1 & \text{if } 3x \geq 2[x] + \left\lceil x + \frac{1}{2} \right\rceil + 1 \end{cases}$$

Case I: $x < [x] + \frac{1}{3}$.

Then, it's also true that $x < [x] + \frac{2}{3}$. Hence,

$$\left\lceil x + \frac{1}{3} \right\rceil = \left\lceil x + \frac{2}{3} \right\rceil = [x] \implies [x] + \left\lceil x + \frac{1}{3} \right\rceil + \left\lceil x + \frac{2}{3} \right\rceil = 3[x]$$

Now, to complete the proof for this case, we must show that $[3x] = 3[x]$.

Observe that $x < [x] + \frac{1}{3} \implies x < [x] + \frac{1}{2} \implies \left\lceil x + \frac{1}{2} \right\rceil = [x]$.

Then, $2[x] + \left\lceil x + \frac{1}{2} \right\rceil = 3[x]$.

And we have $x < [x] + \frac{1}{3} \implies 3x < 3[x] + 1$ so we can conclude that $[3x] = 3[x]$.

This in turn implies that

$$[3x] = 3[x] = [x] + \left\lceil x + \frac{1}{3} \right\rceil + \left\lceil x + \frac{2}{3} \right\rceil$$

Case II: $x \geq [x] + \frac{2}{3}$.

Then, it's also true that $x \geq [x] + \frac{2}{3} \geq [x] + \frac{1}{3}$. Hence,

$$\left\lceil x + \frac{1}{3} \right\rceil = \left\lceil x + \frac{2}{3} \right\rceil = [x] + 1 \implies [x] + \left\lceil x + \frac{1}{3} \right\rceil + \left\lceil x + \frac{2}{3} \right\rceil = 3[x] + 2$$

Now, to complete the proof for this case, we must show that $[3x] = 3[x] + 2$.

Observe that $x \geq [x] + \frac{2}{3} \implies x \geq [x] + \frac{1}{2} \implies [x + \frac{1}{2}] = [x] + 1$.

Then, $2[x] + [x + \frac{1}{2}] = 3[x] + 1$.

And we have $x \geq [x] + \frac{2}{3} \implies 3x \geq 3[x] + 2$ so we can conclude that $[3x] = 3[x] + 2$.

This in turn implies that

$$[3x] = 3[x] + 2 = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right]$$

Case III: $[x] + \frac{1}{3} \leq x < [x] + \frac{2}{3}$.

Then, $[x + \frac{1}{3}] = [x]$ but $[x + \frac{2}{3}] = [x] + 1$, giving us

$$[x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] = 3[x] + 1$$

Now, to complete the proof for this case, we must show that $[3x] = 3[x] + 1$.

There are two sub-cases here.

If $x < [x] + \frac{1}{2}$ then $2[x] + [x + \frac{1}{2}] = 3[x]$ Further, we have $x \geq [x] + \frac{1}{3} \implies 3x \geq 3[x] + 1$ so we can conclude that $[3x] = 3[x] + 1$.

Alternatively, if $x \geq [x] + \frac{1}{2}$ then $2[x] + [x + \frac{1}{2}] = 3[x] + 1$ Further, we have $x < [x] + \frac{2}{3} \implies 3x < 3[x] + 2$ so we can conclude that $[3x] = 3[x] + 1$.

Hence, in each sub-case we have shown

$$[3x] = 3[x] + 1 = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right]$$

□

Solutions are omitted for problems 5 to 8.